

01 DECISION

Commit the first two weeks; preserve later flexibility

63,478

Expected packs across 12 weeks

10,503

Firm first-two-week packing volume

52,980

Remaining rounded base volume:
adjustable

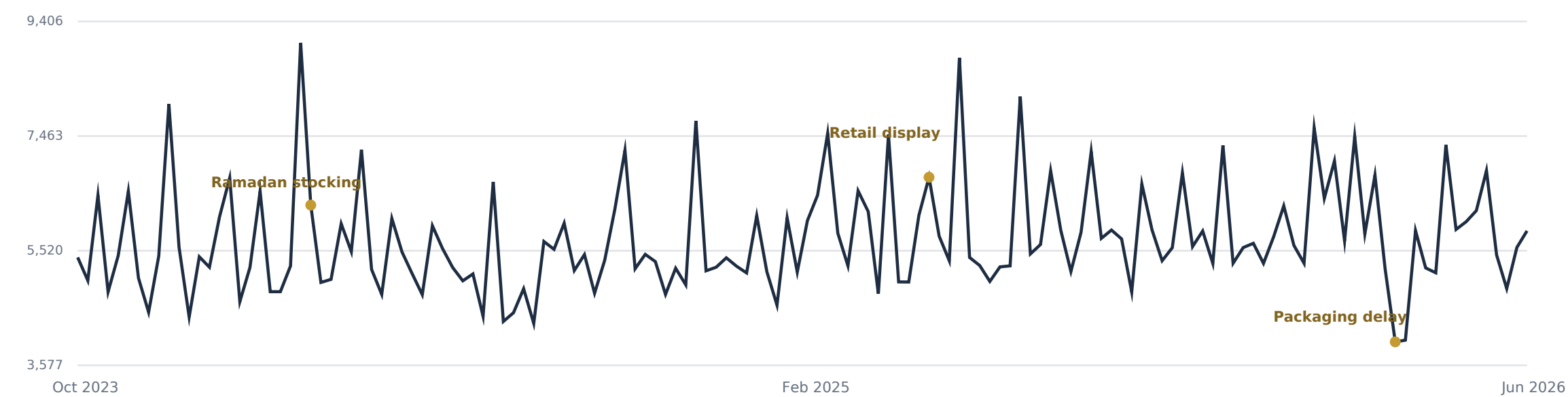
Decision owner: Head of Supply Chain. Plan packing and retail replenishment for the 12-week academic horizon ending 20 Dec 2026.

Why flexibility matters: forecast errors can create excess stock or shortages. The proposal retains a 2,500-pack buffer and reserves 1,000 additional packs/week rather than producing all expected volume now.

Origin: after week ending 27 Sep 2026. Rounded weekly forecast total: 63,483 packs. One unit = one 5 kg pack.

02 DEMAND PATTERN

A stable core is interrupted by promotions and shopping events

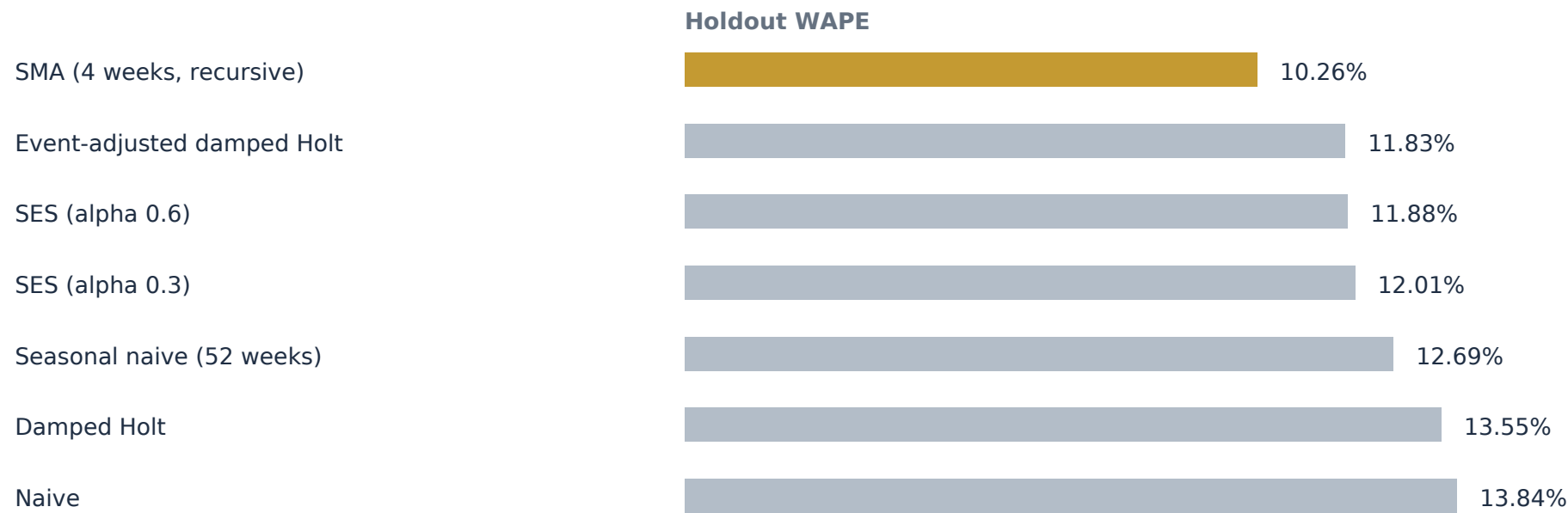


Training mean: 5,715 packs/week. CV: 15.9%. Calendar peaks move between Gregorian weeks; fixed 52-week repeats can misalign Ramadan/Eid.

Promotions and event weeks lift sales in the simulation. Supply constraints suppress recorded sales. Neither the event tags nor the effects represent actual Matco operating history.

03 MODEL CHOICE

The four-week moving average earns a provisional lead



554

MAD packs/week

-2.72

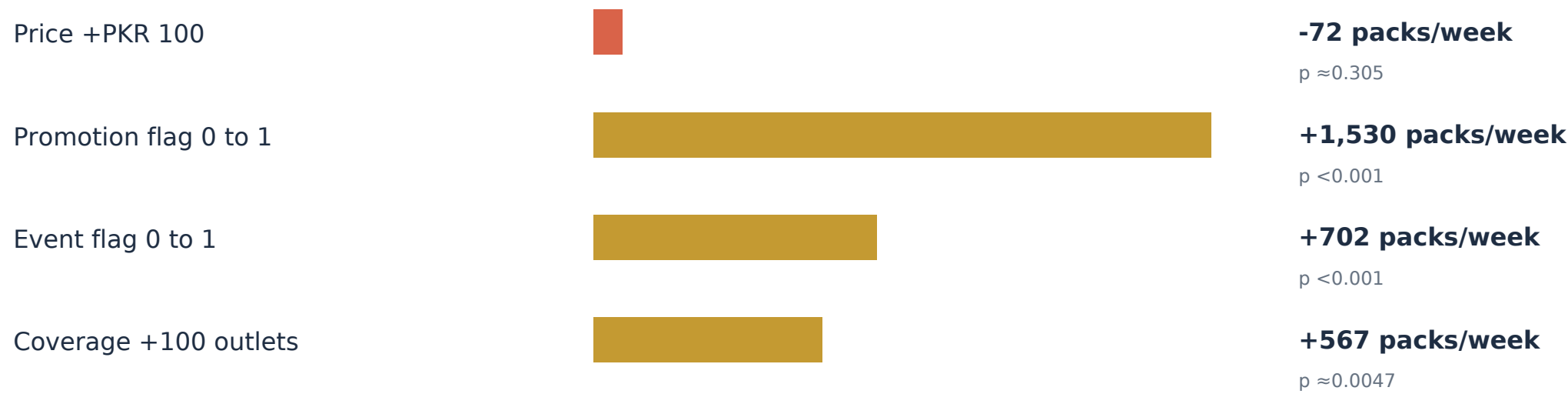
Tracking Signal

Same unseen holdout for all seven methods: 06 Jul–21 Sep 2026. SMA4 also has lowest RMSE (753 packs); bias is -125 packs/week. WAPE improves 25.9% relative to naive.

This short test block has no calendar-event weeks. Event-adjusted Holt and SES 0.6 are close; sophistication alone does not earn selection.

04 DRIVERS

Promotions need supply preparation; pricing evidence is weak



Training R^2 : 0.611; adjusted R^2 : 0.600. Effects hold other included variables constant. P-values shown use approximate four-lag HAC uncertainty.

Operational implication: a promo illustration at PKR 2,865 and 1,106 outlets is about 6,838 packs/week. Reserve a 7,000-pack option and confirm the approved campaign.

Synthetic associations are not causal proof. Price and coverage co-move; omitted discount depth, outlet mix and shocks limit interpretation.

05 RISK

Capacity can absorb uplift; it cannot replace a missing delivery

Largest miss	Error packs	Likely explanation
2026-07-13	+1,560	Promo peak
2026-08-24	-1,456	Stockout scenario
2026-08-03	+941	Promo peak
2026-09-07	-769	Unexplained variation
2026-08-10	-737	Unexplained variation

+20% demand

Regular: 3,171 unfilled packs
Reserved option: 0

Two-week promo shock

Regular: 114 unfilled packs
Reserved option: 0

One-week delivery interruption

Regular: 2,824 unfilled packs
Reserved option: 2,824

The downside –20% scenario peaks at about 4,601 packs in inventory, with assumed stock cost PKR 11.04 million. Reduce later receipts after review.

Capacity, cost, opening buffer and lead time are assumptions. Unfilled volume is not automatically lost sales. Holdout stockouts hide unmet demand.

06 ACTION

Release weekly, reserve capacity and trigger an early review

1. Commit: first two weeks total 10,503 packs (52.52 tonnes). Confirm the 2,500-pack opening buffer and one-week replenishment lead before release.
2. Keep adjustable: baseline 52,980 packs across the remaining ten weeks. Review actuals and inventory each week; avoid converting the full horizon into firm finished stock.
3. Protect supply: verify 6,000 regular packs/week and reserve another 1,000. Activate the option for confirmed uplift above regular capacity. Secure a backup delivery route or available stock for interruptions.
4. Review trigger: two-week combined forecast deviation $>10\%$; rolling eight-week TS outside the assumed ± 4 band; approved campaign above 6,000 packs/week; or a confirmed stockout / receipt interruption.

Owner: Head of Supply Chain. Demand Planning reviews weekly; Sales confirms campaigns; Operations and Procurement confirm stock, capacity and delivery.